Naming convention

Filename grammar

`<scope>_<group>_<interval>.<ext>`

- `scope` identifies the spatial unit.
- `group` identifies the variable family.
- `interval` identifies the time aggregation.
- `ext` is `txt` (always) and `csv` (when `csvout = y` in `print.prt`).

Examples that the source actually opens:

```
hru_wb_day.txt
hru_wb_mon.txt
hru_wb_yr.txt
hru_wb_aa.txt
basin_wb_aa.txt
lsunit_wb_aa.txt
hru-lte_wb_aa.txt
hru_nb_aa.txt
hru_ls_aa.txt
hru_pw_aa.txt
basin_aqu_aa.txt
basin_res_aa.txt
basin_cha_aa.txt
basin_sd_cha_aa.txt
channel_sd_day.txt
channel_day.txt
aquifer_day.txt
reservoir_day.txt
crop_yld_aa.txt
crop_yld_yr.txt
```

Scopes

The scope prefix maps to a writer subroutine:

Scope	Writer	Loop body
`hru_`	`hru_output.f90`	one record per HRU
`hru-lte_`	`hru_lte_output.f90`	one record per HRU-LTE
`lsunit_`	`lsu_output.f90`	one record per landscape unit
`ru_`	`ru_output.f90`	one record per routing unit
`region_`	`lsreg_output.f90`	one record per region per landuse
`basin_`	`basin_output.f90`, `basin_aquifer_output.f90`, `basin_channel_output.f90`, `basin_sdchannel_output.f90`, `basin_reservoir_output.f90`	one record (basin sum or average)
`channel_`	`channel_output.f90`	one record per legacy channel object
`channel_sd` (file group `sd_cha`)	`sd_channel_output.f90`	one record per sd_channel object

Scope	Writer	Loop body
aquifer_	aquifer_output.f90	one record per aquifer
reservoir_	reservoir_output.f90	one record per reservoir
wetland_	wetland_output.f90	one record per wetland
recall_	recall_output.f90	one record per recall object
crop_yld_	hru_output.f90 (last block)	one record per HRU per plant

Groups

The group token identifies the column family. Each writer holds its own Fortran derived type and the columns are the fields of that type in declaration order.

Group	Type	Module
_wb	output_waterbal	src/output_landscape_module.f90
_nb	output_nutbal	src/output_landscape_module.f90
_ls	output_losses	src/output_landscape_module.f90
_pw	output_plantweather	src/output_landscape_module.f90
_cha	ch_output	src/channel_module.f90
_sd_cha	water_body plus two hyd_output records (in, out)	src/water_body_module.f90, src/hydrograph_module.f90
_aqu	aquifer_dynamic	src/aquifer_module.f90
_res	water_body plus two hyd_output records (in, out)	src/water_body_module.f90, src/hydrograph_module.f90
_pest	pesticide processes types	src/output_ls_pesticide_module.f90, others
_salt	salt ion types	src/salt_module.f90
_cs	constituents types	src/cs_module.f90
_carbon, _soilcarb, _rescarb, _plcarb	carbon pool types	src/output_landscape_module.f90, src/carbon_module.f90

Intervals

Suffix	Meaning	print.prt column
_day	one row per day	daily
_mon	one row per month	monthly
_yr	one row per year	yearly
_aa	one row, average annual over the print window	avann

The print window is the simulation range minus nyskip warm-up years (set in the first table of print.prt).

print.prt toggle

print.prt has a table of object groups, one row per group, with four columns (daily, monthly, yearly, avann). Each cell is y or n. The file is opened only if the cell is y. The toggles set pco%<group>%<d|m|y|a> (read in print_read.f90) and the file is opened in output_landscape_init.f90 or one of the header_*.f90 files.

Example: in refdata/Ames_sub1/print.prt, the row

```
hru_wb                    n              n              n              y
```

opens hru_wb_aa.txt only. The other three (day, mon, yr) are not written.

A separate row, csvout, switches on the matching .csv variant for every enabled file.

crop_yld in print.prt accepts n, a (annual only), y (yearly only), or b (both). Other groups use the four-column form.

Subdaily output

When `time%step > 1` (subdaily simulation), additional `*_subday.txt` files are written. The current subdaily channel writer is in `sd_channel_output.f90` (`channel_sd_subday.txt`, unit 2508).

Always-on outputs

A few files are not controlled by `print.prt` and are written every run:

- `simulation.out`. Daily progress log written from `main.f90` and `time_control.f90`.
- `files_out.out`. Index of every output file opened, written from `proc_bsn.f90`.
- `diagnostics.out`. Warnings about missing input references, opened in `proc_bsn.f90`.
- `checker.out`. HRU-level parameter sanity dump, written in `proc_hru.f90`.
- `erosion.out`. USLE inputs and outputs per HRU.
- `success.fin`. Empty marker created on successful completion.

See Checker and diagnostics for the column-level breakdown.